

# **Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?**

Leo Specks

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# **1 Introduction**

## **1.1 Motivation & Background**

## **1.2 The Research Gap**

## **1.3 Research Questions & Hypotheses**

## **1.4 Contributions to the Literature**

## **1.5 Thesis Outline**

## 2 Literature Review

### 2.1 Theories of Commodity Futures Returns

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**Table 1**

UMD Factor: Summary Statistics (Jan 1927–Feb 2026)

| Statistic              | Monthly               | Annualized |
|------------------------|-----------------------|------------|
| Mean Return (%)        | 0.612                 | 7.34       |
| Standard Deviation (%) | 4.68                  | 16.21      |
| Sharpe Ratio           | -                     | 0.453      |
| t-statistic            | 4.51                  | -          |
| p-value (two-tailed)   | $7.11 \times 10^{-6}$ | -          |
| Number of Months       | 1,190                 | -          |

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## 2.2 Aggregate Commodity-Specific Factors

[2]

Guidolin and Pedio [2]

[2]

Guidolin and Pedio

2021

[2, 2]

## 2.3 Individual Commodity Fundamentals

## 2.4 Macroeconomic Predictors

## 2.5 ML Methods in Return Forecasting

## 2.6 Volatility Regimes and Predictability

As you can see in Table 1.

<sup>1</sup>This would be the first footnote

### 3 Methodology

#### 3.1 Baseline Framework (Replication)

We begin by gathering all  $T$  observations of  $n$  variables into a data matrix  $\mathbf{X} \in \mathbb{R}^{T \times n}$ , where **each row is a month** and each column is a macroeconomic variable. Each column is demeaned and standardized to unit variance using the StandardScaler:

$$X_{\text{scaled},ij} = \frac{X_{\text{raw},ij} - \mu_j}{\sigma_j}. \quad (1)$$

Let  $\Sigma = V[\mathbf{X}] \in \mathbb{R}^{n \times n}$  denote the variance-covariance matrix of  $\mathbf{X}$ , which is calculated as:

$$\Sigma = \frac{1}{T-1} \mathbf{X}'\mathbf{X}, \quad (2)$$

and let  $\Sigma = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$  be its spectral decomposition, where  $\mathbf{U} \in \mathbb{R}^{n \times n}$  is an orthogonal matrix of order  $n$ , whose columns are the eigenvectors  $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$  of  $\Sigma$ , and  $\mathbf{\Lambda} \in \mathbb{R}^{n \times n}$  is a diagonal matrix of the same order, whose diagonal entries are the corresponding eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ , which, by convention, are sorted from the largest to the smallest. By multiplying the original scaled data matrix  $\mathbf{X}$  with  $\mathbf{U}$ , we derive the principal components of  $\mathbf{X}$  as the elements of the matrix  $\mathbf{F} \in \mathbb{R}^{T \times n}$ , or formally:

$$\mathbf{F} = \mathbf{X}\mathbf{U}. \quad (3)$$

To assess how much of the total variance in the original  $n$  variables is captured by the  $k$ -th principal component, or alternatively by the first  $K$  principal components, we can compute the variance-covariance matrix of  $\mathbf{F}$  as:

$$\begin{aligned} V[\mathbf{F}] &= V[\mathbf{X}\mathbf{U}] \\ &= \mathbf{U}'\Sigma\mathbf{U} \\ &= \mathbf{U}'\mathbf{U}\mathbf{\Lambda}\mathbf{U}'\mathbf{U} \\ &= \mathbf{I}_n\mathbf{\Lambda}\mathbf{I}_n \\ &= \mathbf{\Lambda}, \end{aligned} \quad (4)$$

where we exploited the orthogonality of  $\mathbf{U}$ . Recalling the fact that  $\mathbf{\Lambda}$  is a diagonal matrix,  $V[\mathbf{F}] = \mathbf{\Lambda}$  implies that the principal components are mutually uncorrelated, and have a variance equal to the associated eigenvalue. Therefore, the proportion of variance in the original  $n$  variables explained by the  $k$ -th principal component is given by:

$$\frac{\lambda_k}{\sum_{i=1}^n \lambda_i}, \quad (5)$$

and the cumulative proportion of explained variance by the first  $K$  components by:

$$\frac{\sum_{k=1}^K \lambda_k}{\sum_{i=1}^n \lambda_i}. \quad (6)$$

Mirroring the approach used by Guidolin and Pedio [2], we forecast the returns of each commodity  $j$ ,  $R_{t+1,j}$ , with the principal component analysis to summarize a large set of macroeconomic factors:

$$R_{t+1,j} = \underbrace{\alpha_j}_{\text{intercept}} + \underbrace{\sum_{i=1}^N \beta_{ij} PC_{t,ij}}_{\text{macroeconomic}} + \underbrace{D_{t,j} \left( \sum_{k=1}^K \gamma_{kj} C_{t,kj} \right)}_{\text{commodity-specific}} + \underbrace{\varepsilon_{t,j}}_{\text{error term}} \quad (7)$$

where  $\alpha_j$  is the model's intercept,

### 3.2 Extension I: Penalized Regression

In the first alternate approach, we keep the principal component analysis and still extract ten principal components from the  $n$  initial variables, but instead of running stepwise AIC on those selected principal components, we run a LASSO regression as an alternative way to further reduce the number of parameters to be estimated. With this only being a narrow replacement compared to the original framework, every difference in results should be attributable solely to the selection method.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{10} \beta_{ij}^{\text{LASSO}} PC_{t,ij} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (8)$$

A broader extension to the approach by Guidolin and Pedio [2] is foregoing the principal component analysis entirely and applying LASSO directly to the  $n$  variables.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{128} \beta_{ij}^{\text{LASSO}} X_{t,i} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (9)$$

### 3.3 Extension II: Individual Fundamental Predictors

### 3.4 OOS Evaluation Criteria

### 3.5 Portfolio Back-Testing Framework



## **4 Data**

### **4.1 Commodity Futures Returns**

### **4.2 Macroeconomic Factors**

### **4.3 Aggregate Commodity-Specific Factors**

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### **5.2 OOS Predictive Accuracy (2004-2012)**

### **5.3 Portfolio Back-Testing (2004-2012)**

### **5.4 Replication Assessment**

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### **6.3 OOS Forecasting Results - Original Sample (2004-2012)**

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### **6.6 Robustness Checks**

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### **7.2 Individual vs. Aggregate Factors: Are They Substitutes?**

### **7.3 OOS Forecasting Results (2004-2024)**

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### **7.6 Regime Interaction**

## **8 Synthesis & Discussion**

### **8.1 Model Horse Race**

### **8.2 Statistical vs. Economic Loss Functions Revisited**

### **8.3 Implications for Commodity Market Segmentation**

### **8.4 Practical Implications**

## **9 Conclusions**

### **9.1 Summary of Findings**

### **9.2 Limitations**

### **9.3 Future Research Directions**

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## References

- [1] Marta Giampietro, Massimo Guidolin, and Manuela Pedio. “Estimating Stochastic Discount Factor Models with Hidden Regimes: Applications to Commodity Pricing”. In: *European Journal of Operational Research* 265.2 (Mar. 1, 2018), pp. 685–702. ISSN: 0377-2217. DOI: 10.1016/j.ejor.2017.07.045.
- [2] Massimo Guidolin and Manuela Pedio. “Forecasting Commodity Futures Returns with Stepwise Regressions: Do Commodity-Specific Factors Help?” In: *Annals of Operations Research* 299.1–2 (Apr. 2021), pp. 1317–1356. ISSN: 0254-5330, 1572-9338. DOI: 10.1007/s10479-020-03515-w.
- [3] Benjamin R. Auer. “Have Trend-Following Signals in Commodity Futures Markets Become Less Reliable in Recent Years?” In: *Financial Markets and Portfolio Management* 35.4 (Dec. 1, 2021), pp. 533–553. ISSN: 2373-8529. DOI: 10.1007/s11408-021-00385-5.
- [4] Massimo Guidolin and Serena Ionta. “Predicting Commodity Returns with Climate Variables: Statistical Loss Functions vs. Economic Value”. In: *Economics Letters* 265 (June 1, 2026), p. 113028. ISSN: 0165-1765. DOI: 10.1016/j.econlet.2026.113028.

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## Appendix

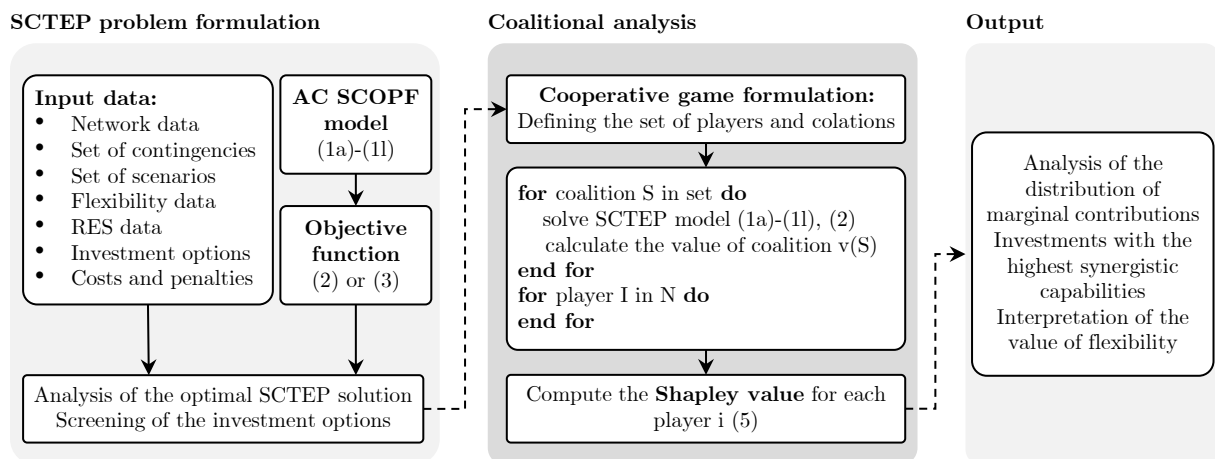
**Table 2**

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| <code>\textcite{}</code>   | Author (Year)  | Author [number]   |
| <code>\parencite{}</code>  | (Author, Year) | [number]          |
| <code>\citeauthor{}</code> | Author         | Author            |
| <code>\citeyear{}</code>   | Year           | Year (not linked) |

**Figure 1**

Example Figure (width should be 16cm)



**Code 1**

Example Codeblock

```
# KPI OVERVIEW
kpis = pd.DataFrame({
    "Ann. Return (approx)": ann_return_simple,
    "Ann. Return (compounding)": ann_return,
    "Ann. Volatility": ann_vol,
    "Sharpe": sharpe,
    "CAGR": cagr,
    "Max Drawdown": max_dd
})

kpis.T
```